K M MOHITH

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SUMMARY

B.Tech Electronics and Communication Engineering student at VIT Chennai (CGPA 9.02/10) with hands-on experience in data analysis, statistical analysis, and building data pipelines and dashboards using Python, SQL, and MATLAB. Research Intern at Samsung R&D Institute India. Experienced in processing large datasets (including 99,754 clinical ECG records), designing controlled experiments with confidence intervals, benchmarking KPIs, and turning results into clear, data-driven recommendations and reports for technical and non-technical stakeholders.

EDUCATION

Vellore Institute of Technology, Chennai **2023 – 2027**

B.Tech in Electronics and Communication Engineering | CGPA: 9.02/10 *Chennai, Tamil Nadu*

Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Computer Networks, Digital Communication, Signal Processing

Marygiri CMI Public School (CBSE) – Class XII: 94.8% **2023**

EXPERIENCE

Research Intern – Samsung PRISM Program, Samsung R&D Institute India **Jul 2025 – Nov 2025**

Bangalore, Karnataka

- Built an end-to-end smart-home device resolution system using YOLOv8n object detection, achieving 99% mAP@50 across seven device classes.
- Implemented a multimodal retrieval (RAG) pipeline using OpenCLIP (ViT-B-32) embeddings stored in a ChromaDB vector database, served through a Python FastAPI backend integrated with Gemini 2.0 Flash for real-time device-state classification and automated control.
- Evaluated model performance using precision, recall, and mAP; benchmarked competing approaches and presented comparative findings to the team during system integration and testing.

Summer Intern – Control & Instrumentation, NSPCL Bhilai (NTPC-SAIL Joint Venture) **May 2025 – Jun 2025**

- Studied the maxDNA Distributed Control System (DCS) of a 2×250 MW thermal power plant and traced data flow from field sensors through remote I/O and controllers to the data historian, alarms, and operator HMI.
- Learned calibration of RTDs, thermocouples, pressure/level transmitters, and 4–20 mA signal loops; documented plant sections and interlock logic in a structured technical report.

PROJECTS

Comparative Performance Analysis of TCP Congestion Control **ns-3.42, FlowMonitor, Statistical Analysis**

- Designed 6 controlled experiments (5–20 Mbps bandwidth, 20–200 ms RTT, 0.5–10× BDP queues, 2–8 flows) comparing NewReno, Cubic, Vegas, and BBR, each run with 5 random seeds and reported with 95% confidence intervals.
- Processed FlowMonitor traces into throughput, delay, RTT inflation, and Jain's fairness index; found BBR cut delay ~20% (68 vs. 85 ms) at 97.8% link utilization with fairness ≥ 0.97, and recommended the best algorithm for each network condition.

OFDM Channel Estimation Framework & Analytics Dashboard **Python, PyTorch, Streamlit, MATLAB**

- Built a Monte Carlo simulation (120 frames × 12 symbols per point, 0–24 dB SNR) benchmarking LS, DFT-LS, MMSE, LMS, and RLS estimators against an ideal baseline using BER, NMSE, and EVM; MMSE reached BER 2.4×10^{-3} with ~12 dB lower NMSE than LS.
- Extended the framework in Python with CNN and MLP estimators and an interactive Streamlit dashboard for model training, data visualization, and real-time comparison of estimators (GitHub, live demo).

ECG Arrhythmia Classification – Healthcare Data Analysis **MATLAB, CNN, Signal Processing**

- Processed 99,754 heartbeats from the MIT-BIH Arrhythmia clinical database through wavelet denoising, normalization, and R-peak segmentation; trained a 1-D CNN classifying five arrhythmia categories at ~88–90% validation accuracy, evaluated with confusion matrices.

TECHNICAL SKILLS

Programming & Databases: Python (NumPy, Pandas), SQL, PostgreSQL, MATLAB, Java, C++, C

Data & Analytics: Data Analysis, Statistical Analysis, Data Visualization, Data Cleaning, Monte Carlo Simulation, Scenario Analysis, Performance Benchmarking, KPI Reporting

Machine Learning & AI: PyTorch, CNN, YOLOv8, OpenCV, RAG, Vector Databases (ChromaDB), LLM Integration

Tools & Frameworks: Streamlit, FastAPI, REST APIs, Git, GitHub, VS Code, ns-3

Core Skills: Problem Solving, Analytical Reasoning, Technical Writing, Presentation, Stakeholder Communication, Team Collaboration

LEADERSHIP & ACTIVITIES

- Technical Team Member, IEEE Photonics Society (VIT Chennai); Organising Committee Member for multiple ideathons and chapter events, coordinating planning, logistics, and participant engagement.